If you are curious about adding solar panels, understand more about bills, and finding out how to evaluate this kind of investment for your home.

Evaluating the Household Economic Impact of Solar Panel Installation

In Southwestern Ontario

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Part 1

I Am Curious About This Whole "Solar" Situation.

I would like to establish, whether installing solar panels in my home would make economic sense and under which conditions it would make sense. The government seems supportive of residential solar installations; however, in my area, the movement appears to be losing momentum. Southwestern Ontario has some of the best solar potential in the country, so I would expect to see far more homes covered in solar panels.

This report is not intended to be a formal research paper. I prefer to think of it as a work in progress—something I will continue writing as I gather more information, test assumptions, and work through the calculations. Perhaps an **economic diary** is the better term. It will develop alongside my understanding of the subject.

The first part is just evaluation of whether the average 2,880 square foot household would see returns back in the foreseeable future and a bit of an analysis on consumption. As many Canadian families saw increase in their electric bills in recent years, I became curious about the viability of solar panels to offset the costs for Canadian households.

Below is a table of the electrical bills by month over the past five years, peaking during the mid-summer months to early fall. Taking this, I take my average monthly bill at $119. This year has been higher than usual. I have also noted there are times where the government has helped with grants, but they are very one-off, temporary and sporadic.

Electric Bills Forecast

We next need to make a rough estimate of the cash flow in investing into a solar panel system. I need to find the cost savings from said installment. My average monthly usage is 690-kWh with on-peak rate at 20.3 cents, mid-peak at 15.7, however the 12 cents is what is advertised as the average cost per customer as per the electrical company website; my bills tell another story, averaging at 15.7 cents. At an average monthly bill of $119, I estimate at $1,428 for this year's bill. This being a very modest estimate. Using an average of anywhere between 3% and 5% growth in the bills forecast. Please note the Independent Electricity System Operator (IESO) states these numbers are closer to 9% in the short-run due to a surge in demand. I have decided to use the 3% and 5% growth scenarios and have projected my bills for the next ten years.

Table 1 – Electricity Rates

Average Monthly bill $119	$ 1,428
Average 690 KWh/month	
On-Peak Rate	0.203
Mid-Peak rate	0.157
Off-Peak rate	0.098
Tier 1 rate	0.120

Figure 1 - Rates as per Electric company

Table 2 – Annual Electricity Bill Forecast for the Next 10 Years

	2026		3% Growth		5% Growth	
Year	Average Monthly Bill (2026)	Yearly Bill (estimate)	Monthly Bill (3%)	Yearly Bill (3%)	Monthly Bill (5%)	Yearly Bill (5%)
2027	$ 119.0	$ 1,428.0	$ 122.6	$ 1,470.8	$ 125.0	$ 1,499.4
2028			$ 126.2	$ 1,515.0	$ 131.2	$ 1,574.4
2029			$ 130.0	$ 1,560.4	$ 137.8	$ 1,653.1
2030			$ 133.9	$ 1,607.2	$ 144.6	$ 1,735.7
2031			$ 138.0	$ 1,655.4	$ 151.9	$ 1,822.5
2032			$ 142.1	$ 1,705.1	$ 159.5	$ 1,913.7
2033			$ 146.4	$ 1,756.3	$ 167.4	$ 2,009.3
2034			$ 150.7	$ 1,808.9	$ 175.8	$ 2,109.8
2035			$ 155.3	$ 1,863.2	$ 184.6	$ 2,215.3
2036			$ 159.9	$ 1,919.1	$ 193.8	$ 2,326.1
2037			$ 164.7	$ 1,976.7	$ 203.5	$ 2,442.4
				$ 18,838.2		$ 21,301.7

Figure 2 - Electricity Bills Forecast Next 10 Years at 3% and 5% Electricity Bill Growth Scenarios.

Cash Flows

I have deducted an additional $180 a year on miscellaneous expenses after the system has been installed. There are three types of sizes of systems used in this analysis, as a 7-kW system seems capable of covering the energy needs by 100% and a 6-kW will still give a surplus. That will be evaluated later on. But for now, a 5-kW system will provide a 2% Internal Rate of return over eleven years. Remember the payout works as long as the bills are higher in the future. My estimates are very conservative at 5%.
A 5-kW system is estimated at $17,500, this is a rough estimate, and it includes all costs included, paperwork, installation, etc.[1]

[1] https://solarcalculatorcanada.org/solar-ontario

Table 3 – IRR for 7, 6 and 5-Kilowatt Systems with a $180 cost of Maintenance

	7-kW Solar Panel System		6-kW Solar Panel System		5-kW Solar Panel System	
YEAR	GROWTH 3%	GROWTH 5%	GROWTH 3%	GROWTH 5%	GROWTH 3%	GROWTH 5%
	25,000	25000.0	21000	21,000	17,500	17,500
Year 1	-1470.8	-1499.4	-1470.8	-1499.4	-1470.8	-1499.4
Year 2	-1515.0	-1574.4	-1515.0	-1574.4	-1515.0	-1574.4
Year 3	-1560.4	-1653.1	-1560.4	-1653.1	-1560.4	-1653.1
Year 4	-1607.2	-1735.7	-1607.2	-1735.7	-1607.2	-1735.7
Year 5	-1655.4	-1822.5	-1655.4	-1822.5	-1655.4	-1822.5
Year 6	-1705.1	-1913.7	-1705.1	-1913.7	-1705.1	-1913.7
Year 7	-1756.3	-2009.3	-1756.3	-2009.3	-1756.3	-2009.3
Year 8	-1808.9	-2109.8	-1808.9	-2109.8	-1808.9	-2109.8
Year 9	-1863.2	-2215.3	-1863.2	-2215.3	-1863.2	-2215.3
Year 10	-1919.1	-2326.1	-1919.1	-2326.1	-1919.1	-2326.1
Year 11	-1976.7	-2442.4	-1976.7	-2442.4	-1976.7	-2442.4
IRR	**-4%**	**-2%**	**-2%**	**0%**	**1%**	**3%**

Figure 3 – Internal Rate of Return for an Ordinary Cash Flow for 7, 6 and 5-kW Systems with $180 Maintenance and Assuming 3% and 5% Annual Electricity Bill Growth.

Note, I will not be seeing my money back in most scenarios, at least over the next ten years.

Historical Use and Evaluation of Usage

The next leg of the evaluation takes into consideration the approximate output of a Solar panel, in southwestern Ontario. The output of a solar panel during the summer is 6.32 kW a day per Kilowatt, a 6-kW system outputs a total of 1,137.6 units for 30 days.

Table 4 – Expected Output by Season of a 6 and 5-Kilowatt System in Southern Ontario

	Kilowatt/Day	Days/Mth	6 kW System		5 kW System	
SUMMER	6.32	30	6	1137.6	5	948.0
FALL	3.19	30	6	574.2	5	478.5
WINTER	1.98	30	6	356.4	5	297.0
SPRING	5.38	30	6	968.4	5	807.0

Figure 4 - Expected Output for Solar Panel System (IESO).

The following table shows our historical monthly electricity consumption for the past six years. I will calculate the average monthly consumption and subtract the expected solar energy production to determine our net electricity usage.

Our net electricity consumption will vary depending on the season. During the summer months, a 6-kW solar system is expected to generate more electricity than we consume, making us a net

contributor to the grid. Based on our calculations, a 6-kW system would produce an annual net surplus of approximately 768.4 kWh. In contrast, a 5-kW system would significantly reduce our electricity bills but would still leave us with an annual net consumption of approximately 760 kWh. These results suggest that the ideal system size lies somewhere between 5 and 6 kW to achieve net-zero annual electricity consumption.

At this stage, our analysis does not account for the costs associated with purchasing, installing, operating, or maintaining the solar system. These financial considerations will be addressed in Part Two of our analysis.

Table 5 – Historical Monthly Usage of Electricity Total KWH/Month

Month	2026	2025	2024	2023	2022	2021	Average
Jan	566	550	531	512	545	537	**540**
Feb	566	576	510	468	354	539	**502**
Mar	648	521	471	479	602	399	**520**
Apr	512	492	473	479	513	457	**488**
May	533	512	402	469	571	425	**485**
Jun	468	460	978	827	923	899	**759**
Jul	858	1,097	1,211	1,144	1,190	1,010	**1,085**
Aug	1,027	1,272	1,166	1,182	1,274	1,117	**1,173**
Sep	1,095	937	1,073	1,000	1,079	1,052	**1,039**
Oct	905	923	767	682	608	724	**768**
Nov		512	468	462	473	478	**479**
Dec		566	524	466	493	465	**503**
Total	**7,178**	**8,418**	**8,574**	**8,170**	**8,625**	**8,102**	**8,178**

Figure 5 - Historical Monthly Electricity Usage.

Table 6 – Expected Output per Year of a 6-Kilowatt System vs a 5-Kilowatt System

Month	Expected Output from a 6-kW System	Difference (Expected Output Solar - Average Usage) 6-kW System	Expedted Output from a 5-kW system	Difference (Expected Output Solar - Average Usage) 5-kW System
Jan	356	(184)	297	(243)
Feb	356	(146)	297	(205)
Mar	968	448	807	287
Apr	968	481	807	319
May	968	483	807	322
Jun	1,138	378	948	189
Jul	1,138	53	948	(137)
Aug	1,138	(35)	948	(225)
Sep	574	(465)	479	(561)
Oct	574	(194)	479	(290)
Nov	574	96	479	(0)
Dec	356	(146)	297	(206)
Total	**9,110**	**768**	**7,592**	**(750)**

Figure 6 - Expected Output - Average Usage of a 6-Kilowatt vs a 5-Kilowatt System.

Part 2.

All the Nitty Gritty Details About Southern Ontario and Quotes from Real Contractors

Alright, for this section we have already obtained some information and are a bit more sober on our expectations. I got an actual quote from a local solar panel system supplier and installer, the quote I got was of $30,000 CDN with a turn key service. Very different from the $17,500 advertised on the Ontario Solar Power website. I also found out there's a fixed delivery fee from the power company for residential consumption of $35 a month. Regardless of having solar or not, call it a "membership" fee. I also found out I can install at most a 6-kW system as I don't have the space. He also commented that these systems are solid for the very first 10 years. There is a 25-year warranty on panels, 15 years on inverters and five on labor.

So, after *this* bit of information, I removed the $180 maintenance cost per year as stated previously but I will add the $35 delivery fee, which adds up to $420 per year. The result is that we break-even after 20 years under the 3% yearly increase in prices scenario and 16 years under the 5% price scenario.

Shortly after, I got a quote from another company; but this time, for a 5-kW system for a price of $23,000. Under this scenario I will break even in year 16 under the 3% price appreciation scenario and year 14 under the 5% scenario. With a 4% discount rate, the only scenario where we come out green is in the 5-kW system with a 5% expected bill growth.
Another issue that I had not considered, was that I have a ten-year old roof, and I got advised to update the side where the panelling would have to be installed. Another expense, which I have *not* factored into our calculations.

Table 7 – Projected 20-Year Cash Flows and Internal Rate of Return (IRR) for a 6 and 5-kW Systems with a $35 per Month Fixed Surcharge, Assuming a 3% and 5% Electricity Bill Growth

6-kW Solar Panel System with $35 Monthly Fixed Delivery Fee			5-kW Solar Panel System with $35 Monthly Fixed Delivery Fee		
	GROWTH 3%	GROWTH 5%		GROWTH 3%	GROWTH 5%
	30,000	30,000		23,000	23,000
Year 1	(1,051)	(1,079)	Year 1	(1,051)	(1,079)
Year 2	(1,095)	(1,154)	Year 2	(1,095)	(1,154)
Year 3	(1,140)	(1,233)	Year 3	(1,140)	(1,233)
Year 4	(1,187)	(1,316)	Year 4	(1,187)	(1,316)
Year 5	(1,235)	(1,403)	Year 5	(1,235)	(1,403)
Year 6	(1,285)	(1,494)	Year 6	(1,285)	(1,494)
Year 7	(1,336)	(1,589)	Year 7	(1,336)	(1,589)
Year 8	(1,389)	(1,690)	Year 8	(1,389)	(1,690)
Year 9	(1,443)	(1,795)	Year 9	(1,443)	(1,795)
Year 10	(1,499)	(1,906)	Year 10	(1,499)	(1,906)
Year 11	(1,557)	(2,022)	Year 11	(1,557)	(2,022)
Year 12	(1,616)	(2,144)	Year 12	(1,616)	(2,144)
Year 13	(1,677)	(2,273)	Year 13	(1,677)	(2,273)
Year 14	(1,740)	(2,407)	Year 14	(1,740)	(2,407)
Year 15	(1,805)	(2,549)	Year 15	(1,805)	(2,549)
Year 16	(1,872)	(2,697)	Year 16	(1,872)	(2,697)
Year 17	(1,940)	(2,853)	Year 17	(1,940)	(2,853)
Year 18	(2,011)	(3,017)	Year 18	(2,011)	(3,017)
Year 19	(2,084)	(3,188)	Year 19	(2,084)	(3,188)
Year 20	(2,159)	(3,369)	Year 20	(2,159)	(3,369)
IRR	0%	3%	IRR	3%	5%

Figure 7 - Projected 20-Year Cash Flows and IRR for a 6 and 5-kW Systems with a $35 per Month Fixed Surcharge, Assuming a 3% and 5% Electricity Bill Growth.

OK, I Need to Add Depreciation Expenses Too, No?

Depreciation is such a mixed bag and I will provide a good breakdown for anyone interested in this subject. Panel degradation is expected at 0.4% per year. The expected lifespan of the complete system is at 25 years. Now, what I was actually looking for is replacement risk, i.e., the chance a component creates a future cash expense. It is calculated as follows:

Replacement risk = probability of failure x out of pocket replacement cost.

Now, the panel system is less risky but: 25 years is a very long time and the technology might be obsolete, the company out of business, electrical standards might have changed. That being said, the above formula is used for the panel maintenance: 11 panels, with 0.1% probability of failure, at $600 replacement cost per panel (we consider the additional cost of hiring someone, etc.). For a span of 20 years, this gives a value of $132.

The inverter is the major replacement risk, with one replacement approximately mid-way in the 25-year lifespan. I will put the present value of an inverter at year 10 replacement into the initial cost of the system at $1,750.

The rest of the maintenance I estimate at $100 per year, not $180 as in the beginning of the analysis. This is really just a cover for anything that might arise over the span of the system. Remember this value is more likely to be zero for the first few years of the installation.

Expected panel repairs ($6.60 * 20) = $132.
Expected inverter replacement = $1,750 + $1,750 (electrician) = $3,500.
Routine maintenance ($100 * 20) = $2,000.
Total expected 20-year expenses = $3,882.
Present value at 4% discount rate = **$2,408**.

A Little More Refinement…

Since I did take into consideration the costs of a 5-KWh system simply because I don't have the space in my roof for anything larger, I *should* refine my analysis. For starters a 5-KWh system does NOT cover all my expenses, unlike the 6-KWh system. I will have to refine in such a way that it will reduce my savings projections as it gives me a net usage of 750-KW per year, at 12 cents it gives $90 per year.

The contractor mentioned a local company called Canadian Solar. I went on their website and found they sell panels at various price levels and can cost $160 per panel[2]. I would need 11 panels; my main hardware stack looks like this:

Hardware:

1. Canadian Solar panels: **~$1,650–$1,950.**
2. Solis 5 KW inverter: **~$1,850–$2,340.**
3. Roof racking/attachments: **~$900–$1,200.**
4. Rapid-shutdown devices: **~$550–$700.**
5. Wiring, conduit, breaker, disconnect, grounding, labels, monitoring: probably **~$1,000–$2,000.**

[2] https://www.santansolar.com/product/new-grade-b-canadian-solar-525w-bifacial-solar-panel/

Our total cost for parts now goes from lower **$5,950** to upper **$8,190**. So, it seems the main price increase is in labour, and certifications, inspections, etc.

Permits and fees:

Town permit: $80.
ESA inspection/notification: $336.
Drawings/design: $300–$600.
Structural review: $0–$750.
Utility meter/connection: $500–$1,000 provisional allowance.
Labels/documentation/miscellaneous: $150–$300.

Ok, this makes it from **$1,366** to **$3,066**.

Now for the labour:

Roof crew/mechanical installation: ~$1,500–$2,500.
Licensed electrician/solar electrical work: ~$1,000–$2,000.
Total labour: roughly **$2,500–$4,500**.

Everything together can go for lower end between **$9,856** and upper end **$15,756**. This is much better than the **$23,000** I was quoted for turnkey. I will produce one last set of two spreadsheets, with the different initial costs and the below additional costs into the investment.

Factors affecting the Project Investment return:

- **Installation:** Scenarios of $9,856, $15,756 and $23,000 today.
- **Electricity price increases:** 3% and 5% yearly cost increases estimates.
- **Fixed delivery fee:** of $35 a month, $420 per year.
- **Yearly production vs savings:** Actual solar kW × actual avoidable $/kW.

- **Inverter:** middle case ~50% × $3,500 replacement, discounted from around year 11; Or conservative case assumes the full replacement (used this scenario).
- **Maintenance and panel repairs:** $2,132 (as calculated in previous section).
- **Discount Rate:** 4%.

The first half of Table 9 displays the bill forecast for projected savings at a 3% average electricity bill, which is considered very conservative. The second column is the fixed $35 electricity disposal fee, so that is not a saving and must be removed from our savings. The third column is the net usage, as a 5-Kilowatt system is not covering all the expenses, must be added as well.

We have decided to add the second half of the costs into the cashflow column directly, the inverter was added at $3,500 during year 14, assuming complete replacement and paid in full during that year. That price should also cover labour. Maintenance and panel repairs have been accounted each year at $100 and $6.6 spread out during the 21-year IRR calculation.

We can see below for a cash flow of 3% average annual increase in electricity prices, over a twenty-one-year span, we get an Investment Rate of Return of 7%, breaking even with 0% at year 11 (yellow), and 4% (in blue) in year 16, which is comparable to the discount rate. Under the cheapest scenario of an initial expenditure of $9,856. It would be safe to conclude the starting cost should be somewhere between $9,856 and $15,756 and break even after the 16th year of the investment.

Table 8 - Projected 20-Year Cash Flows and Internal Rates of Return (IRR) for a 5-kW Solar Panel System, Assuming 3% Annual Electricity-Bill Growth and 4% Discount Rate

5-kW Solar Panel System				GROWTH 3%		
	Bill Forecast (at 3% Growth)	Fixed fee (35*12)	Net usage of 750kWh/year	9,856	15,756	23,000
Year 1	(1,428.0)	420	112.5	(789)	(789)	(789)
Year 2	(1,470.8)	433	115.9	(816)	(816)	(816)
Year 3	(1,515.0)	446	119.4	(843)	(843)	(843)
Year 4	(1,560.4)	459	122.9	(872)	(872)	(872)
Year 5	(1,607.2)	473	126.6	(901)	(901)	(901)
Year 6	(1,655.4)	487	130.4	(932)	(932)	(932)
Year 7	(1,705.1)	502	134.3	(963)	(963)	(963)
Year 8	(1,756.3)	517	138.4	(995)	(995)	(995)
Year 9	(1,808.9)	532	142.5	(1,028)	(1,028)	(1,028)
Year 10	(1,863.2)	548	146.8	(1,062)	(1,062)	(1,062)
Year 11	(1,919.1)	564	151.2	(1,097)	(1,097)	(1,097)
Year 12	(1,976.7)	581	155.7	(1,133)	(1,133)	(1,133)
Year 13	(2,036.0)	599	160.4	(1,170)	(1,170)	(1,170)
Year 14	(2,097.1)	617	165.2	2,292	2,292	2,292
Year 15	(2,160.0)	635	170.2	(1,248)	(1,248)	(1,248)
Year 16	(2,224.8)	654	175.3	(1,289)	(1,289)	(1,289)
Year 17	(2,291.5)	674	180.5	(1,330)	(1,330)	(1,330)
Year 18	(2,360.3)	694	185.9	(1,374)	(1,374)	(1,374)
Year 19	(2,431.1)	715	191.5	(1,418)	(1,418)	(1,418)
Year 20	(2,504.0)	736	197.3	(1,464)	(1,464)	(1,464)
Year 21	(2,579.1)	759	203.2	(1,511)	(1,511)	(1,511)
IRR				7%	2%	-1%

Figure 8 - Projected 21-Year Cash Flows and Internal Rates of Return (IRR) for a 5-kW Solar Panel System, Assuming 3% Annual Electricity-Bill Growth. In Blue: Net 4% IRR = 4% Discount Rate.

For the 5% scenario, things look a bit more optimistic. The break-even is in year 10 (in yellow) for the lowest price case scenario and 16 for the middle case. However, comparing with the discount of 4% it would be between Year 12 and 20. That is, if I had my money in the bank at 4%, I would be breaking even with this investment somewhere between the 13th, and 20th year assuming that my total system installation costs were somewhere between $9,856 and the $15,756 price. The turn-key scenario does not give any breakeven, at least before year 20. The rate of return for all three scenarios is: 9%, 4% and 1% respectively. $23,000 for a 5-kW system, even at 5% appreciation in electricity prices is *not* a good deal.

Table 9 - Projected 20-Year Cash Flows and Internal Rates of Return (IRR) for a 5 kW Solar Panel System, Assuming 5% Annual Electricity-Bill Growth and 4% Discount Rate

5-kW Solar Panel System						
				GROWTH 5%		
	Bill Forecast (at 5% Growth)	Fixed fee (35*12)	Net usage of 750kWh/year	9,856	15,756	23,000
Year 1	(1,428.0)	420	112.5	(789)	(789)	(789)
Year 2	(1,499.4)	441	118.1	(834)	(834)	(834)
Year 3	(1,574.4)	463	124.0	(881)	(881)	(881)
Year 4	(1,653.1)	486	130.2	(930)	(930)	(930)
Year 5	(1,735.7)	511	136.7	(982)	(982)	(982)
Year 6	(1,822.5)	536	143.6	(1,036)	(1,036)	(1,036)
Year 7	(1,913.7)	563	150.8	(1,093)	(1,093)	(1,093)
Year 8	(2,009.3)	591	158.3	(1,153)	(1,153)	(1,153)
Year 9	(2,109.8)	621	166.2	(1,216)	(1,216)	(1,216)
Year 10	(2,215.3)	652	174.5	(1,283)	(1,283)	(1,283)
Year 11	(2,326.1)	684	183.3	(1,352)	(1,352)	(1,352)
Year 12	(2,442.4)	718	192.4	(1,425)	(1,425)	(1,425)
Year 13	(2,564.5)	754	202.0	(1,502)	(1,502)	(1,502)
Year 14	(2,692.7)	792	212.1	1,918	1,918	1,918
Year 15	(2,827.3)	832	222.7	(1,666)	(1,666)	(1,666)
Year 16	(2,968.7)	873	233.9	(1,755)	(1,755)	(1,755)
Year 17	(3,117.1)	917	245.6	(1,848)	(1,848)	(1,848)
Year 18	(3,273.0)	963	257.9	(1,946)	(1,946)	(1,946)
Year 19	(3,436.7)	1,011	270.7	(2,049)	(2,049)	(2,049)
Year 20	(3,608.5)	1,061	284.3	(2,156)	(2,156)	(2,156)
Year 21	(3,788.9)	1,114	298.5	(2,269)	(2,269)	(2,269)
IRR				**9%**	**4%**	**1%**

Figure 9 - Projected 21-Year Cash Flows and Internal Rates of Return (IRR) for a 5 kW Solar Panel System, Assuming 5% Annual Electricity-Bill Growth. In Blue: Net 4% IRR = 4% Discount Rate.

Conclusion

I can see why solar panel systems are not as popular as I imagined. They are still very steep as an upfront cost, and it's a shame as Southwestern Ontario is very sunny. The turn-key scenario is very expensive. The best option would be to shop for all the panels and other parts in person, find the biggest deals and shop around for labor.